SDL Lessons 1

SDL_CreateWindow:-

Link : https://wiki.libsdl.org/SDL2/SDL_CreateWindow
`SDL_CreateWindow` is a function in the SDL2 library used to **create a window** on the screen.

Basic Syntax:

```
SDL_Window * SDL_CreateWindow(const char *title,
                              int x, int y, int w,
                              int h, Uint32 flags);
```

Examples:

```
//defining window
       //syntax screen title ,x and y position on opening ,than screen
with and height
       //than
       window = SDL_CreateWindow(
           "SDL Lesson1",
           SDL_WINDOWPOS_UNDEFINED,
           SDL_WINDOWPOS_UNDEFINED,
           SCREEN_WIDTH,
           SCREEN_HEIGHT,
           SDL_WINDOW_SHOWN);
```

Function Parameters

const char *	**title**	the title of the window, in UTF-8 encoding.
int	**x**	the x position of the window, `SDL_WINDOWPOS_CENTERED`, or `SDL_WINDOWPOS_UNDEFINED`.
int	**y**	the y position of the window, `SDL_WINDOWPOS_CENTERED`, or `SDL_WINDOWPOS_UNDEFINED`.
int	**w**	the width of the window, in screen coordinates.
int	**h**	the height of the window, in screen coordinates.
Uint32	**flags**	0, or one or more SDL_WindowFlags OR'd together.

Remarks of Flags

flags may be any of the following OR'd together:

- SDL_WINDOW_FULLSCREEN: fullscreen window
- SDL_WINDOW_FULLSCREEN_DESKTOP: fullscreen window at desktop resolution
- SDL_WINDOW_OPENGL: window usable with an OpenGL context
- SDL_WINDOW_VULKAN: window usable with a Vulkan instance
- SDL_WINDOW_METAL: window usable with a Metal instance
- SDL_WINDOW_HIDDEN: window is not visible
- SDL_WINDOW_BORDERLESS: no window decoration
- SDL_WINDOW_RESIZABLE: window can be resized
- SDL_WINDOW_MINIMIZED: window is minimized
- SDL_WINDOW_MAXIMIZED: window is maximized
- SDL_WINDOW_INPUT_GRABBED: window has grabbed input focus
- SDL_WINDOW_ALLOW_HIGHDPI: window should be created in high-DPI mode if supported (>= SDL 2.0.1)

`SDL_WINDOW_SHOWN` is ignored by SDL_CreateWindow(). The SDL_Window is implicitly shown if SDL_WINDOW_HIDDEN is not set. `SDL_WINDOW_SHOWN` may be queried later using SDL_GetWindowFlags().

SDL_GetWindowSurface:

https://wiki.libsdl.org/SDL2/SDL_GetWindowSurface

Syntax

```
SDL_Surface * SDL_GetWindowSurface(SDL_Window * window);
```

SDL_Window *	**window**	the window to query.

Example:

```
surface = SDL_GetWindowSurface(window);
```

A new surface will be created with the optimal format for the window, if necessary. This surface will be freed when the window is destroyed. Do not free this surface.

This surface will be invalidated if the window is resized. After resizing a window this function must be called again to return a valid surface.

Note that on some platforms the pixels pointer of the surface may be modified after each call to SDL_UpdateWindowSurface(), so that the platform code can implement efficient double or triple buffering.

```
SDL_UpdateWindowSurface(window);

This updates the window after the surface
```

SDL_fill REACT:

🔴 What this line does (overall)

👉 It **fills a surface with a color**
👉 In this case: **solid RED color**

Defined in SDL_surface.h

SYNTAX:

SDL_FillRect(SDL_Surface* dst, SDL_Rect* rect, Uint32 color);

```
SDL_FillRect(surface, NULL, SDL_MapRGB(surface->format, 255, 0, 0));
```

What does NULL MEAN here:

NULL

- This means:
  👉 "Fill the **entire surface**"

If you passed a rectangle:

SDL_Rect r = {0, 0, 100, 100};
SDL_FillRect(surface, &r, color);

👉 Only that area would be filled

SDL_MapRGB

Defined in SDL_pixels.h

```
Uint32 SDL_MapRGB(const SDL_PixelFormat * format,
                  Uint8 r, Uint8 g, Uint8 b);
```

```
    //Surface define
            surface = SDL_GetWindowSurface(window);
            SDL_FillRect(surface, NULL, SDL_MapRGB(surface->format, 255,
0, 0));
            SDL_UpdateWindowSurface(window);
```